MAT1575, Classwork13, Fall2026

$a_1=5,\ a_n=a_{n-1}+2$

5.1 Sequences (p.374-390) $\{a_n\}: a_n=3+2n,\ a_1=3+2=5,\ a_2=7,\ \ldots\ a_2=7,\ a_3=9$ terms

1.*Definition*: **Terminology of Sequences**

Infinite sequence $\{a_n\}$: an ordered list of numbers with index value n and they are

Explicit formulas: $a_n=f(n)$ for some function f(n) defined over positive integers.

Recurrence Relation: one or more terms are given, and subsequent terms are defined in terms in the sequence

2. *Examples*.

(a) **Arithmetic Sequence**: Given $a_1=3, a_n=a_{n-1}+4$ for $n\geq 2$. Find the explicit formula for a_n.

$a_1=3,\ a_2=a_1+4=7,\ a_3=a_2+4=11,\ a_4=a_3+4=15\ \cdots\ a_5=a_1+4\cdot 4$

$=a_1+4+4$

$=a_1+2\cdot 4$

$=a_1+4+4+4$

$=a_1+3\cdot 4$

$\Rightarrow a_n=a_1+(n-1)\cdot 4$

In general, $a_1=3$ is the first term, $d=a_n-a_{n-1}=$ common difference $a_n=3+(n-1)\cdot 4$

(b) **Geometric Sequence**: Given $a_1=2, a_n=-\frac{1}{3}\cdot a_{n-1}$ for $n\geq 2$. Find the explicit formula for a_n.

In general,

3. *Examples*: Find the Limit of each Sequence when $n\to\infty$.

(a)$\{a_n\}=\{3n+1\}$:

(b)$\{b_n\}=\left\{1-\left(\frac{1}{2}\right)^n\right\}$:

(c)$\{c_n\}=\{(-1)^n\}$:

(d)$\{d_n\}=\left\{\frac{(-1)^n}{n}\right\}$:

4. *Definition*: **Convergent and Divergent Sequence**.

Given a sequence $\{a_n\}$. If the term a_n become , we say

$\{a_n\}$ is a and L is the limit of the sequence: .

If $\{a_n\}$ is not convergent, we say it is

5. **Algebraic Limit Laws**. Given sequences $\{a_n\},\{b_n\}$ and a real number c. If there exist finite numbers A and B such that $\lim_{n\to\infty} a_n=A,\ \lim_{n\to\infty} b_n=B$.

(a) $\lim_{n\to\infty} c=$

(b) $\lim_{n\to\infty} ca_n=$

(c) $\lim_{n\to\infty}(a_n\pm b_n)=$

(d) $\lim_{n\to\infty}(a_n\cdot b_n)=$

(e) $\lim_{n\to\infty}\left(\frac{a_n}{b_n}\right)=$

6. *Examples*: For each given sequence, determine whether or not the sequence converge and find its limit.

(a) $\left\{\frac{3n^4-7n^2+5}{6-4n^4}\right\}$

(b) $\left\{\frac{2^n}{n^2}\right\}$

(c) $\left\{\left(1+\frac{4}{n}\right)^n\right\}$

7. **Squeeze Theorem** for Sequences.

Consider sequences $\{a_n\}, \{b_n\}$, and $\{c_n\}$. Suppose there exists an integer N and a finite number L such that

for all $n \geq N$ and , then $\{b_n\}$ and

8. Use the Squeeze Theorem to find the limit of each given sequence:

(a) $\left\{\frac{\cos(n)}{n^2}\right\}$ (b) $\left\{\left(-\frac{1}{2}\right)^n\right\}$

9. *Definition*: **Convergent and Bounded Sequences**. Given a sequence $\{a_n\}$ and a real number M.

(a) Bounded above: $\{a_n\}$ is

(b) Bounded below: $\{a_n\}$ is

(c) $\{a_n\}$ is a bounded sequence if it is

(d) If $\{a_n\}$ is not a bounded sequence, it is an unbounded sequence.

(e) If $\{a_n\}$ converges, then it is bounded.

10. *Definition*: **Monotone Sequence**. Given a sequence $\{a_n\}$.

(a) $\{a_n\}$ is increasing for all $n \geq n_0$ if

(b) $\{a_n\}$ is decreasing for all $n \geq n_0$ if

(c) $\{a_n\}$ is a monotone sequence for all $n \geq n_0$ if

11. **Monotone Convergence Theorem**.

Monotone Convergence Theorem
If $\{a_n\}$ is a bounded sequence and there exists a positive integer n_0 such that $\{a_n\}$ is monotone for all $n \geq n_0$, then $\{a_n\}$ converges.